

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

**Duration : 1 Hour
Total Marks:30**

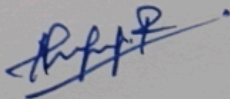
Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I R. Akshaya (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:





1. A school must create a timetable assigning 3 subjects (Math, Science, English) to 3 periods (Period 1, Period 2, Period 3) such that:

Constraints:

- i. The Math teacher is unavailable during Period 3
- ii. Science must be scheduled before English (Period 1 < Period 2 < Period 3)
- iii. Only one subject per period
- iv. English cannot be scheduled in Period 1
- v. Every subject must be assigned exactly one period

Apply Backtracking Search to find a valid timetable, showing all intermediate steps and constraint checks. State the final valid schedule.

2. An autonomous rescue drone is deployed in a disaster-affected building to locate a survivor. The building is represented as a grid of rooms, and the drone operates in a partially observable environment where some paths are blocked and some are risky.

Constraints:

- i. The drone can move Up, Down, Left, Right
- ii. Some cells are blocked (obstacles) and cannot be traversed
- iii. The drone has limited sensing (can only observe adjacent cells)
- iv. Each movement has a cost of 1; entering risky zones costs an additional 2
- v. The drone must minimize total cost while ensuring safe navigation
- vi. The drone must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze the constraints and explain how they affect the drone's decision-making.
- b) Develop a solution using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent), explaining the steps involved.
- c) Demonstrate how the drone applies AI concepts such as intelligent agents, rationality, and environment type.
- d) Justify why your chosen approach is most suitable under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

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1. SCHOOL TIME TABLE USING BACKTRACKING SEARCH.

Step 1: Variables

* Subjects = {Math, Science, English}

* Periods = {P₁, P₂, P₃}

Step 2: Constraints

1. Math cannot be in Period 3
2. Science must be before English.
3. Only one subject per period.
4. English cannot be in Period 1.
5. Every subject must be assigned Exactly one period.

Step 3: Backtracking Process.

Attempt 1:

* P₁ = Math, Remaining: Science, English.

* P₂ = Science

* P₃ = English.

Constraint Check:

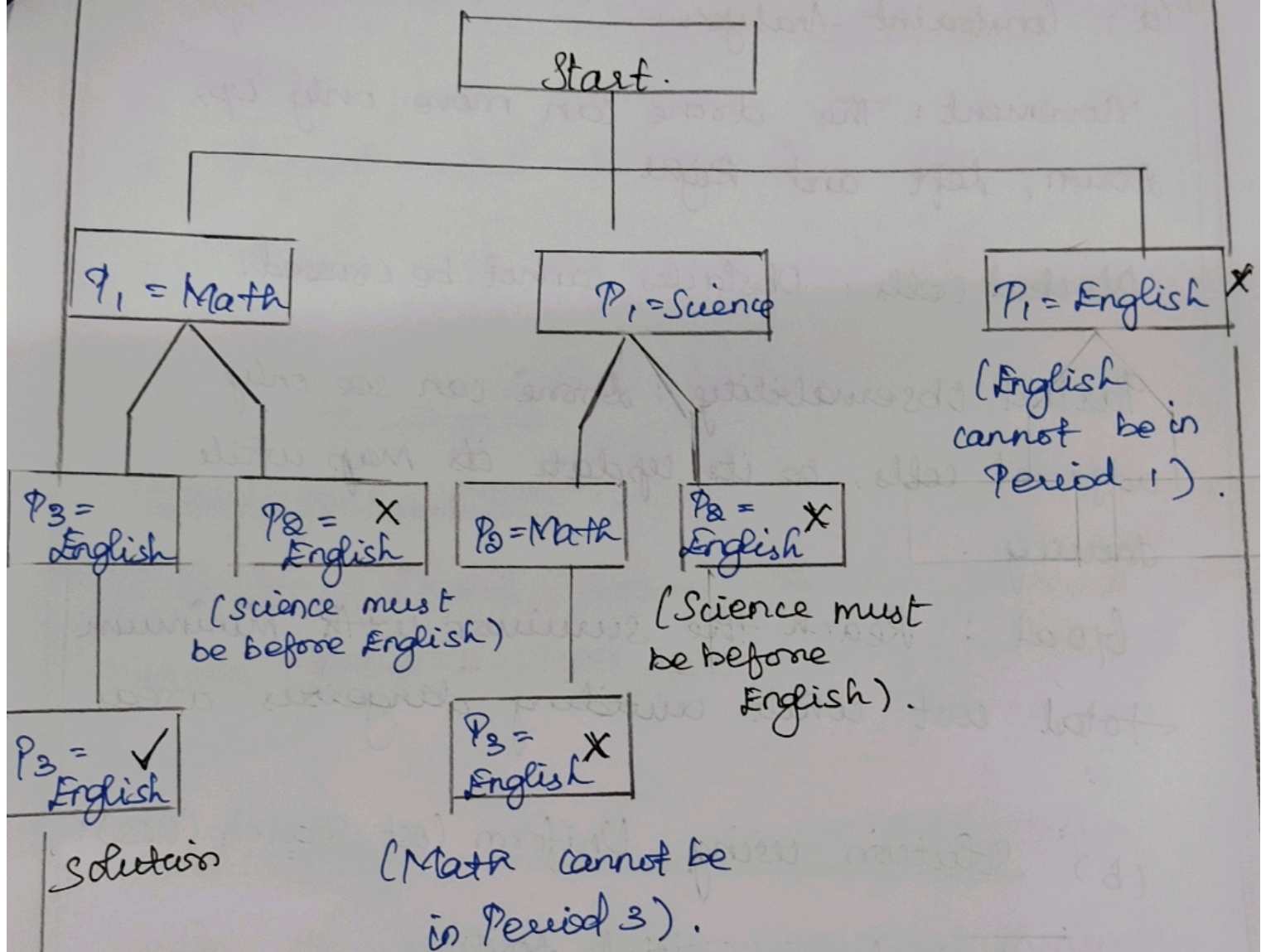
- * Math not in P_3
- * Science before English
- + One subject per period
- * English not in P_1
- * All subjects assigned.

Since all constraints are satisfied, no further backtracking is required.

Final Valid Timetable:-

Period	Subject.
Period 1	Math
Period 2	Science
Period 3.	English

Backtracking Search Tree :-



Conclusion :

First assignment satisfies all constraints,
the final timetable is :

Period 1 $\rightarrow$ Math

Period 2 $\rightarrow$ Science

Period 3 $\rightarrow$ English.

2. Autonomous Rescue Drone:

(a) Constraint Analysis:

Movement: The drone can move only Up, Down, Left and Right.

Blocked cells: Obstacles cannot be crossed.

Partial Observability: Drone can see only adjacent cells, so it updates its map while moving.

Goal: Reach the survivor with minimum total cost while avoiding dangerous areas.

(b) Solution using Uniform Cost Search (UCS):

1. Start from initial position.
2. Insert the start node into the priority queue with cost = 0
3. Expand the node with lowest cumulative cost
4. Ignore blocked cells.

5. Add movement cost (1 for normal, 3 for risky).
6. Update priority queue with newly discovered nodes.
7. Repeat until the survivor reached.
8. Return the minimum-cost path.

(c) AI Concepts:

Intelligent Agent:

⇒ The drone senses the Environment through sensors and acts using movement controls.

Rationality:

⇒ It chooses the action that minimizes total travel cost while safely reaching the survivor.

Environment Type:

* Partially Observable: Cannot see the whole map.

- * Dynamic : Environment changes as new areas explored.
- * Sequential : Every action affects future decisions.
- * Deterministic : Moving in direction produces a predictable result.

(d) Justification :

- ⇒ It guarantees the minimum cost path
- ⇒ Handles different movement cost effectively.
- ⇒ Avoids risky zones unless necessary.
- ⇒ Ensures safe and efficient rescue operations.

Conclusion :

The rescue drone behave as rational intelligent agent, using Uniform cost search to safely navigate a partially observable Environment while minimizing total travel cost.